

Name Class Date **P4B - Chapter 4 Quiz**

Total questions: 5

Worksheet time: 6mins

1. **PROC SQL;**
SELECT *
FROM accounts a, rates r
WHERE a.type= r.type;
QUIT;

Based on the code of the image, what is the type of join being done?

- a) INNER JOIN b) RIGHT JOIN
c) LEFT JOIN d) FULL JOIN

2. **DATA merged;**
MERGE accounts rates;
BY type;
RUN;

Based on the code of the image what will be the output after execution?

(Note: The accounts and rates dataset are not sorted.)

- a) SAS will automatically sort the datasets before merging b) The code will run correctly but slower
c) SAS will produce an error because datasets must be sorted d) SAS will produce a warning and results may be incorrect

3. Two datasets are being merged by the variable **customer_id** in **SAS**. In the first dataset, each **customer_id** appears only once, while in the second dataset some **customer_id** values appear multiple times. Additionally, there are some **customer_id** values that exist in the first dataset but do not exist in the second dataset.

This situation represents a **one-to-many merge**, and the observations whose **customer_id** appears only in the first dataset will still be included in the final merged dataset with missing values for the variables coming from the second dataset.

- a) True b) False

4.

```
DATA detailed_purchases;
  SET PURCHASES;
  BY client_id;
  IF first.client_id THEN first_pur = amount;
  IF last.client_id THEN last_pur = amount;
RUN;
```

Consider an SAS dataset called purchases with 3 variables (client_id, purchase_date and amount), already sorted by **client_id** and **purchase_date**.

Based on the code of the image, what will the detailed_purchases dataset contain for the variables first_pur and last_pur?

- a) first_pur contains the cumulative total of purchases per client; last_pur contains the last purchase amount.
- b) Both first_pur and last_pur are missing for all observations.
- c) first_pur contains the last purchase amount, and last_pur contains the first purchase amount for each client.
- d) first_pur contains the first purchase amount for each client, and last_pur contains the last purchase amount for each client.

5.

```
PROC SORT data=clients;
  BY client_id;
Run;
PROC SORT data=purchases;
  BY client_id;
Run;

DATA merged;
  MERGE clients(in=inC) purchases(in=pur);
  BY client_id;
  IF inC;
Run;
```

Which of the following **PROC SQL** queries produces the **same result** as the DATA step above?

a)

```
PROC SQL;
  CREATE TABLE merged AS
  SELECT *
  FROM clients
  INNER JOIN purchases
  ON clients.client_id = purchases.client_id;
QUIT;
```

b)

```
PROC SQL;
  CREATE TABLE merged AS
  SELECT *
  FROM clients
  RIGHT JOIN purchases
  ON clients.client_id = purchases.client_id;
QUIT;
```

c)

```
PROC SQL;
  CREATE TABLE merged AS
  SELECT *
  FROM clients
  LEFT JOIN purchases
  ON clients.client_id = purchases.client_id;
QUIT;
```

d)

```
PROC SQL;
  CREATE TABLE merged AS
  SELECT *
  FROM clients
  FULL JOIN purchases
  ON clients.client_id = purchases.client_id;
QUIT;
```

Answer Keys

1. a) INNER JOIN
2. c) SAS will produce an error because datasets must be sorted
3. a) True
4. d) first_pur contains the first purchase amount for each client, and last_pur contains the last purchase amount for each client.
5. c)